

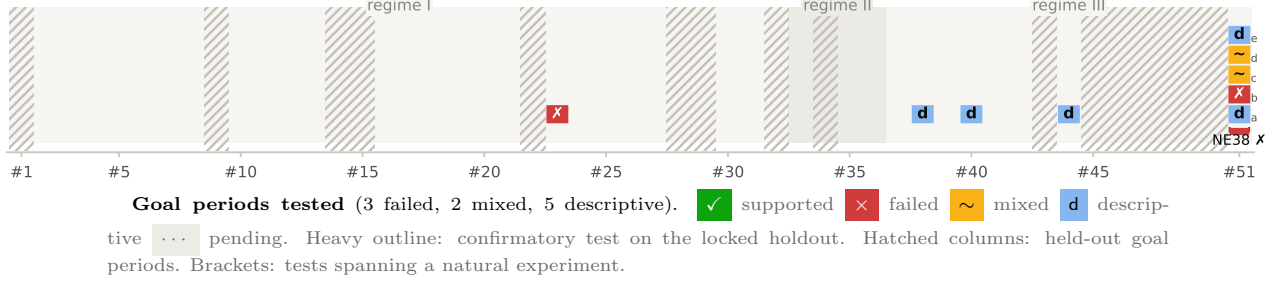
H22: Private, conflicting goals make #51 a spin glass

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

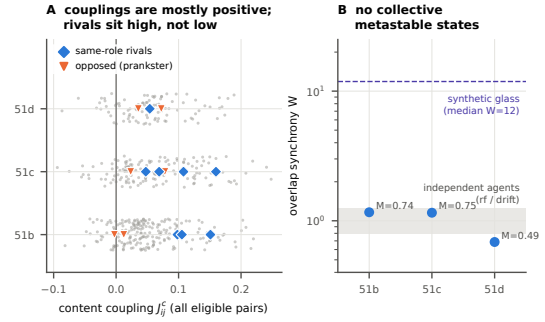
Bottom line. #51 is not a spin glass: private roles make it a random-field system with weak, mostly positive coupling in both topic and stance, where rivals align rather than repel and the swarm never switches collectively.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Do #51's private, partly conflicting roles make the swarm a spin glass? Tested on 51a-e (45 non-holdout days) against shared-objective weeks #38, #40, #44. Content couplings are real in 51b and 51c ($\rho_{\text{split}} = 0.46$, $p = 0.003$) but mostly positive; the content balance index is indeterminate; positions persist without collective switching ($W \approx 1$). Same-role rivals co-move *more* than unrelated pairs ($T_{\text{SR}} = +0.048$, $p = 0.02$). **Round 1b** (DQ6 roles incl. Opus 5's recovered role, two embedding models, style residuals, dedupe, corrected talk, shared unit splits, a DQ2 stance channel) keeps this in every variant ($T_{\text{SR}} +0.04$ to $+0.06$, $p \leq 0.05$). Stance is ferromagnetic (mean 0.6) with unbalanced residuals ($\tau_3^{\text{dc}} 0.05\text{--}0.25$) but no hostility between rivals or opposed roles. Natives: Opus 5's role change and chess opponents show no antagonism. Holdout **not run**.



Observables. (a) Every eligible pair's content coupling in the counted #51 units (gray), with same-role rivals (blue) and Prankster vs Ethicist/Psychologist pairs (orange): mostly positive, rivals high. (b) Overlap synchrony W at the independent-agent level (gray band), far below a synthetic glass (dashed). High memory M without synchrony: agents hold their own positions; the swarm never jumps between collective states.

PHYSICAL MODEL

Models 01 (SK) and 11 ($O(n)$ spins). Spin $\mathbf{s}_i$: agent i 's whitened chat embedding per 30-min window ($n = 32$). Fields: day field $\mathbf{m}_d$; static field $\mathbf{H}_i$ (role, lab, style). J_{ij} : correlation of within-day fluctuations $\mathbf{x}_i, \mathbf{x}_j$ minus a cross-day surrogate; A, B, C : day folds. κ : SK ratio J_0/J ; τ_3 : gauge-invariant balance (SK ≈ 0 , factions ≈ 0.5); q : overlap of day-demeaned states $\hat{\delta}$; W : its synchrony beyond per-agent shifts.

$$P(\{\mathbf{s}\}) \propto \exp \left(\sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (\mathbf{m}_d + \mathbf{H}_i) \cdot \mathbf{s}_i \right)$$

$$J_{ij} = \text{corr}_w(\mathbf{x}_i, \mathbf{x}_j) - \text{corr}_w^{\text{surr}}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\kappa = \sqrt{N} \bar{J} / \sigma_J, \quad \tau_3 = \text{tr}(J^A J^B J^C) / \text{tr}(J^A J^B)^{3/2}$$

$$q(d, d') = \frac{1}{N} \sum_i \hat{\delta}_{i,d} \cdot \hat{\delta}_{i,d'}$$

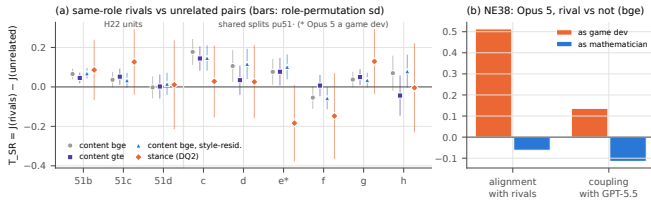
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Practically, little. "Spin glass" was the exciting reading of mixed-motive swarms: frustrated, history-dependent, hard to steer. The data say something duller and more usable. Assigned roles act as fixed fields that pin each agent's topic, the shared room adds a weak positive pull, and same-objective agents drift together. For an operator, roles, not inter-agent conflict, set what such a swarm talks about, and there are no hidden collective states to escape. Caveat: embeddings measure topic, not stance, so strategic opposition could be invisible here. The diagnostics port to other logs; the spin-glass vocabulary is not earned.

H22-private-goals-spin-glass · generated 2026-10-04 10:55 by infra/summaries/build_summaries.py · card hypotheses/H22-private-goals-spin-glass/README.md · ratings: RE-C2 round-1b agent (suggested; faithfulness and usefulness unchanged from the coordinator's calibration)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (51b / 51c / Verdict 51d)
P1 real couplings	content 2/3 → 2/3 (bge), 3/3 (gte); ✓ stance 3/3 beyond agent fields
P2 $\kappa < 1$	5.2 / 3.9 / 9.7; stance mean +0.6 ×
P3 $\tau_3^{\text{dc}} < 0.25$	content indeterminate (both models); stance stance 0.25 / 0.23 / 0.05 ✓
P4 rivals repel	RE $T_{\text{SR}} + 0.048 \rightarrow +0.042$ (gte), +0.052 × (ri- (style-res.); stance +0.09 n.s. val)
P4 opposed roles	content ≈ 0 ; stance +0.11 / -0.34 / × +0.22 (2 pairs)
P5 collective states	W 1.16 / 1.15 / 0.69; gte 51c 1.61 ($p \times$ 0.02)
Talk (secondary)	noise floor → heterogeneous on the fixed new table; $T_{\text{SR}} \approx 0$
NE38 (native)	field leaves rivals (p 0.004); coupling DiD × -0.27 n.s.
G23 chess (na- tive)	opponents' stance -0.005; content × +0.03 n.s.



Second figure. (a) Rival contrast T_{SR} per unit in content (two models and style-residualized) and stance; bars: role-permutation sd. Positive almost everywhere; the one negative stance value (pu51e, where Opus 5 is a game dev) is not significant. (b) NE38: Opus 5's alignment with and coupling to its game-dev rivals while it was one of them, and after.

GOAL PERIODS IN DETAIL

G23 × failed
G38 d 1b: descriptive (round 1: descriptive)
G40 d 1b: descriptive (round 1: descriptive)
G44 d 1b: descriptive (round 1: descriptive)
G51a d 1b: descriptive (round 1: descriptive)
G51b × 1b: descriptive (round 1: failed; rival homophily in every...
G51c ~ 1b: inconclusive (round 1: inconclusive)
G51d ~ 1b: inconclusive (round 1: inconclusive)
G51e d 1b: descriptive (round 1: descriptive)
NE38 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

τ_3^{dc} cannot separate SK from a heterogeneous ferromagnet (Amendment 1), so the unbalanced stance residuals are not glass evidence. Per-pair negative-stance counts assume replies independent within a pair, so threads inflate them; they are not enriched in conflict classes. Few testable rival pairs (3–5 per unit; one in most shared splits). The NE38 coupling result uses a threshold chosen after seeing the coverage. Stance labels are reliable only in aggregate (pair-level “opposes” precision 0.08). Holdout untouched.

ROUND 2

Lagged (kinetic) couplings to separate the room’s common drive from coupling; a reply-level stance model with role-time covariates.

- **Essence.** Do conflicting incentives produce conflict behavior at all?
- **H22-R1** Cooperation default: agents help rivals at rates comparable to aligned pairs (Jev-labeled help acts).
- **H22-R2** A quantitative random-field fit (HH129): roles as fields plus weak positive coupling.